🧠 AI Model Accuracy: House Price Predictions – Student Activity

🎯 Objective:

Students will calculate the accuracy of an AI model by analyzing how well it predicted house prices. They will apply math formulas (absolute difference, percentage, and average) and understand how we evaluate AI performance in real-world problems.

📋 Instructions:

1. **Look at the table** provided. The AI has predicted house prices, and we are comparing them with the actual market prices.
2. **Use the formulas** in the first row to calculate the missing values for:
    - **Error (Absolute difference)** = |Actual Price - Predicted Price|
    - **Error Rate** = Error / Actual Price
    - **Accuracy** = 1 - Error Rate
    - **Accuracy %** = Accuracy × 100
3. **Fill in the table** row by row using your calculator or mental math.
4. At the end, **find the mean of all Accuracy values** to get the **AI Model's Final Accuracy**.
5. Discuss: **Is the AI model good at predicting prices? Why or why not?**

📝 Student Worksheet Table (Fill in the blanks)

Predicted House Price (USD)	Actual House Price (USD)	Error Abs (Actual - Predicted)	Error Rate (Error / Actual)	Accuracy (1 - Error Rate)	Accuracy % (Accuracy × 100)
391k	402k	Abs(402k - 391k) = 11k	11k / 402k = 0.027	1 - 0.027 = 0.973	97.3%
453k	488k				
125k	97k				
871k	907k				
322k	425k				

Helpful Formula Box (for students to refer to):

- **Error** = |Actual - Predicted|
- **Error Rate** = Error ÷ Actual
- **Accuracy** = 1 - Error Rate
- **Accuracy %** = Accuracy × 100
- **Mean Accuracy** = Sum of all Accuracy values ÷ Number of Samples

Note: <u>Name the worksheet worksheet (Excel) with your Name and Roll Number</u>